

Natural Language Processing

(Background and Overview)

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Course Organization

Recommended Reading:

- *Speech and Language Processing*, Book by Daniel Jurafsky and James H. Martin
- Steven Bird, Ewan Klein, Edward Loper - Natural Language Processing with Python

Evaluation Criteria :

80% paper, 20% Assignments

Assignments:

- Minimum 2 Assignments (50% for Each)

Course Organization

- Introduction to NLP
- Words and Tokens
- Data Collection and Text Pre-Processing
- Text Representation
- CFG, POS Tagging and NER
- Introduction to Deep Learning
- Language Modeling and LLM, Pre-Trained LLMs
- Machine Translation
- Speech Recognition and Text to Speech Systems
- Natural Language Generation/Summarization
- Evaluation of NLP Applications

What is NLP?

Natural Language VS Programming Language

- Not another programming language.
- Natural languages are what humans use to share information with each other. It has developed naturally in use.
- A computer program written with a programming language tells a machine exactly what to do.
- But there are no compilers or interpreters for natural languages.

Natural Language

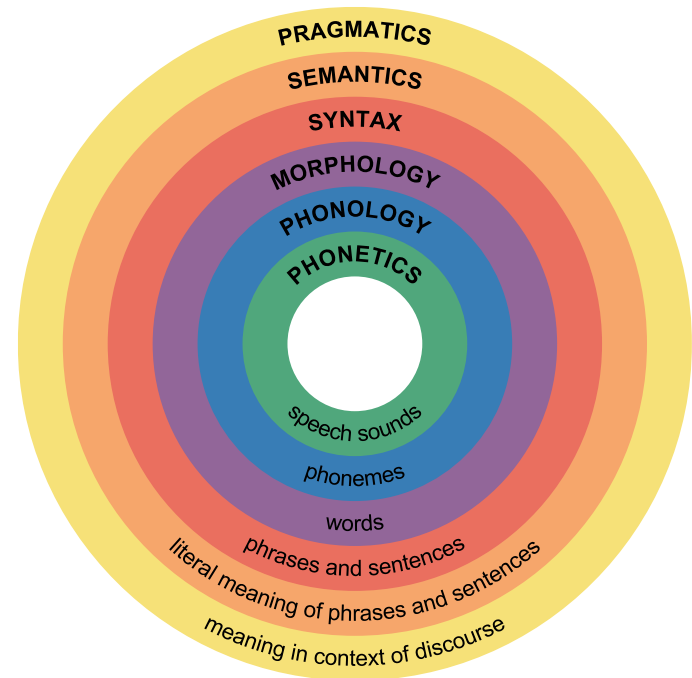
- Natural language refers to the spoken and written forms of communication that humans use every day to convey information, meaning, and emotion.

NLP

- Natural Language Processing (NLP) is a subfield of artificial intelligence (AI) that focuses on enabling computers to understand, interpret, and generate human language in a way that is meaningful and useful.
- NLP combines computational linguistics, which models the structure of human language, with machine learning and deep learning models to process and analyze large amounts of natural language data.

Linguistics

- The scientific study of language and its structure, including the study of grammar, syntax, and phonetics.
 - Specific branches of linguistics include sociolinguistics, dialectology, psycholinguistics, computational linguistics, comparative linguistics, and structural linguistics.



Importance of NLP

- **Enhanced Human-Computer Interaction:**

NLP allows for more natural ways for humans to interact with computers.

Eg: voice commands, chatbots, and virtual assistants, etc.

Importance of NLP

- **Insights from Text Data:** NLP helps in extracting valuable insights from unstructured text data.
- Eg: customer reviews, social media posts, and news articles.
- This can inform business decisions, monitor public opinion, and enhance user experiences.

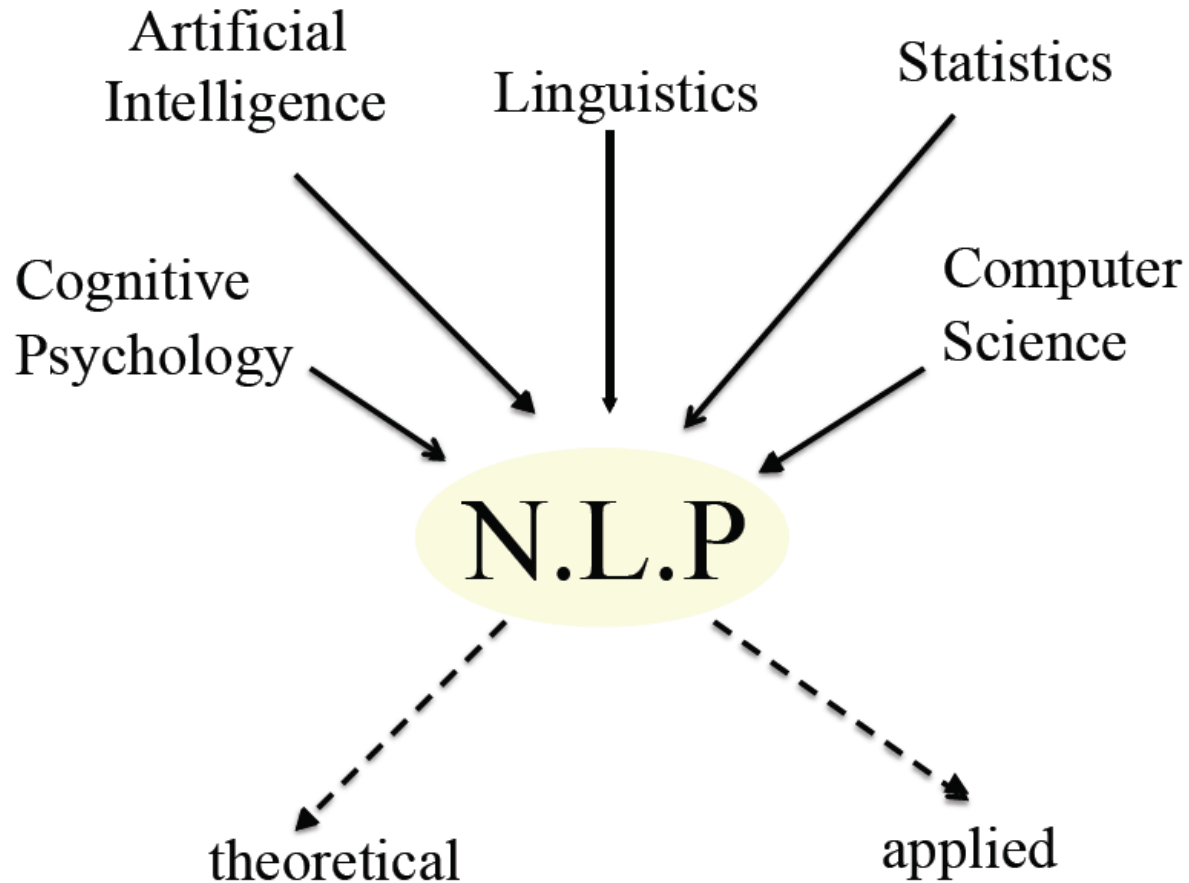
Importance of NLP

- **Multilingual Communication:** NLP enables the translation of text and speech from one language to another, breaking down language barriers and facilitating global communication.

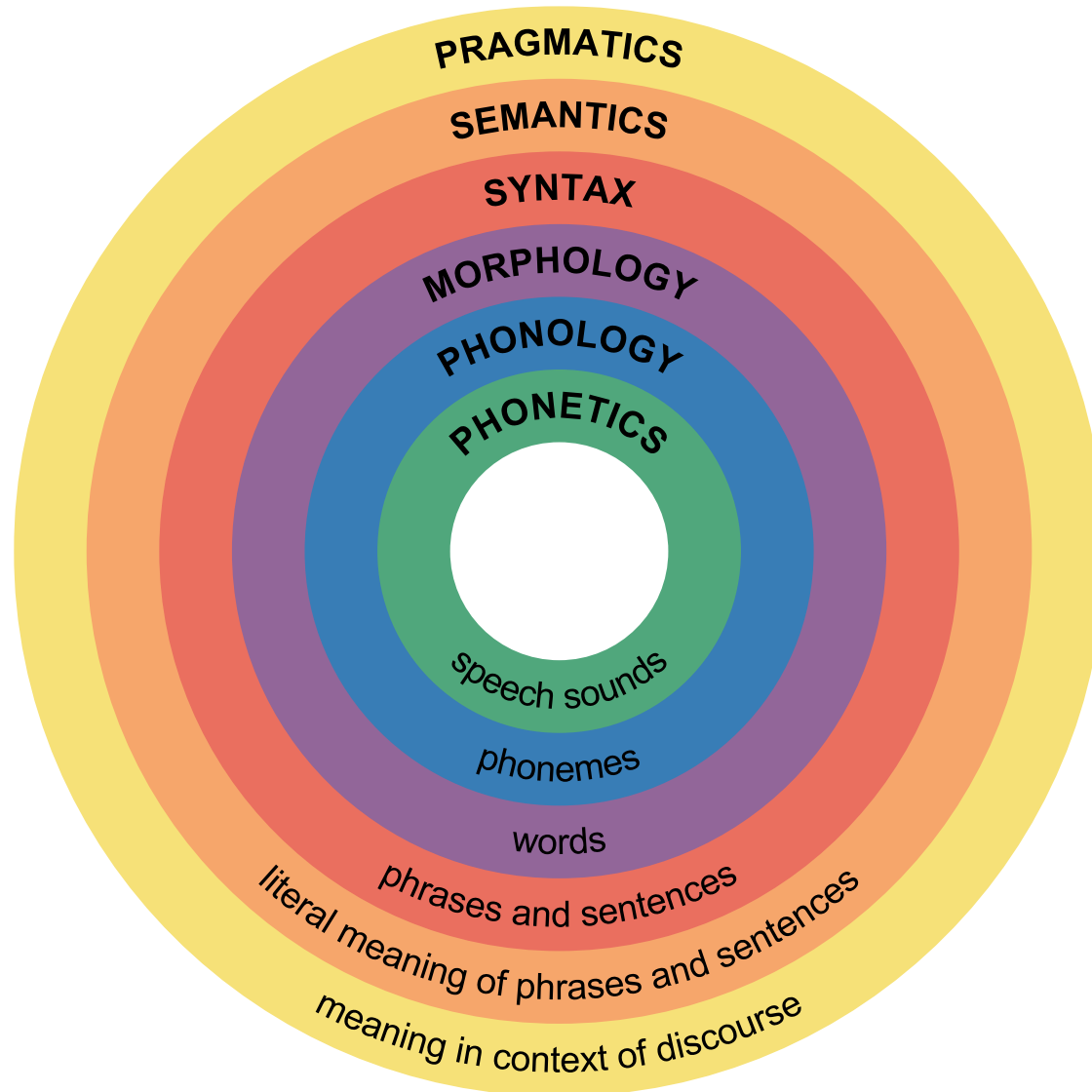
Importance of NLP

- **Information Retrieval:** NLP improves search engines and information retrieval systems, making it easier to find relevant information quickly and efficiently.

NLP



Different levels of linguistic analysis



The levels of linguistic analysis are:

- **Phonetics and Phonology** - the study of sounds
- **Morphology** - the study of word components
- **Syntax** - the study of sentence and phrase structure
- **Lexical Semantics** - the study of the meanings of words
- **Compositional Semantics** - how to combine words
- **Pragmatics** - how to accomplish goals
- **Discourse conventions** - how to deal with units larger than utterances

The levels of linguistic analysis are:

(1) **Phonetics and Phonology** - The study of the sound systems of languages and the rules for their combination.

- NLP Application:
 - Speech Recognition
 - Text-to-Speech (TTS)

The levels of linguistic analysis are:

(2) **Morphology** - The study of the structure and formation of words, including morphemes, which are the smallest units of meaning.

– NLP Application:

- **Tokenization**: Breaking down text into words and morphemes.
- **Lemmatization and Stemming**: Reducing words to their base or root forms.

The levels of linguistic analysis are:

(3) **Syntax** - The study of the rules and principles that govern the structure of sentences.

– NLP Application:

- **Part-of-Speech Tagging**: Identifying the grammatical categories of words.
- **Parsing**: Analyzing the grammatical structure of sentences.

The levels of linguistic analysis are:

(4) Lexical Semantics - Study the meaning of individual words and the relationships between words.

Word meaning

bank → can mean

- financial institution
- river bank

The levels of linguistic analysis are:

(5) **Compositional Semantics** - Study how the meanings of words combine to form the meaning of phrases and sentences.

John loves Mary.

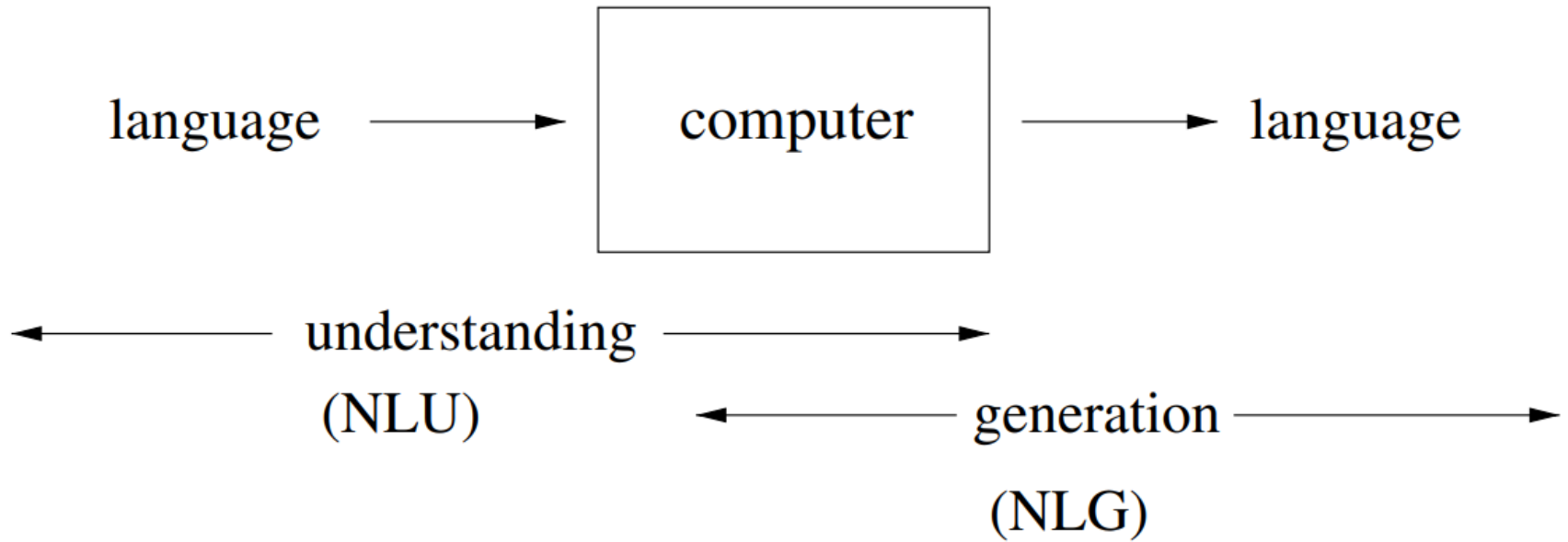
Mary loves John.

The same words appear in both sentences, but the structure changes the meaning.

Natural Language Understanding and Natural Language Generation

NLU and NLG

- **NLU** – Natural Language Understanding
 - NLU helps the machine to understand and analyse human language by extracting the metadata from content such as concepts, entities, keywords, emotion, relations, and semantic roles.
- **NLG** – Natural Language Generation
 - NLG acts as a translator that converts the computerized data into natural language representation. It mainly involves Text planning, Sentence planning, and Text Realization.



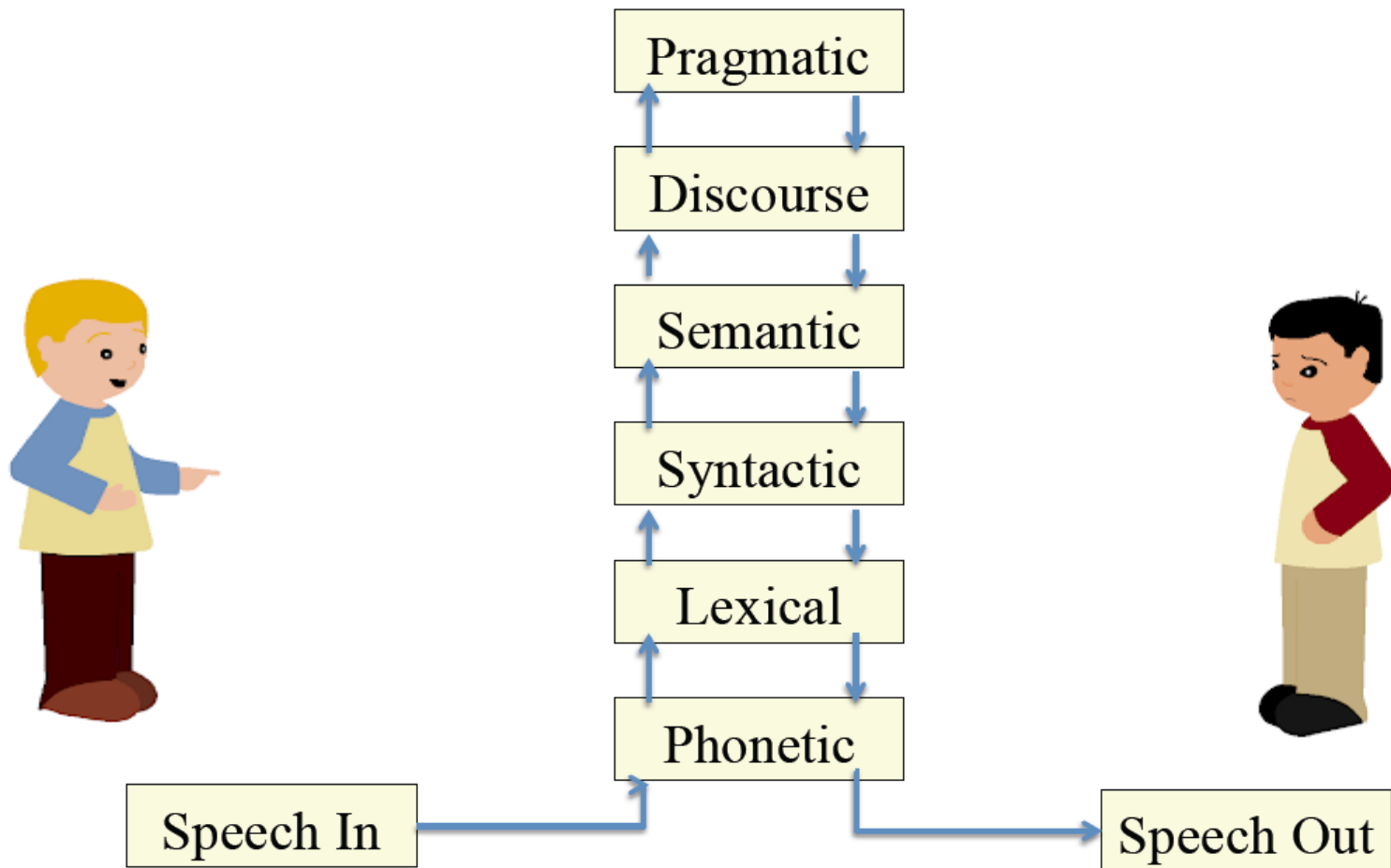
- Computers using natural language as input and/or output

NLU and NLG

– Applications:

- Voice assistants like Alexa or Siri that respond to commands.
- Machine translation tools that translate one language into another (automated translation services)
- Conversational AI assistants

A Linear Model of Language Processing



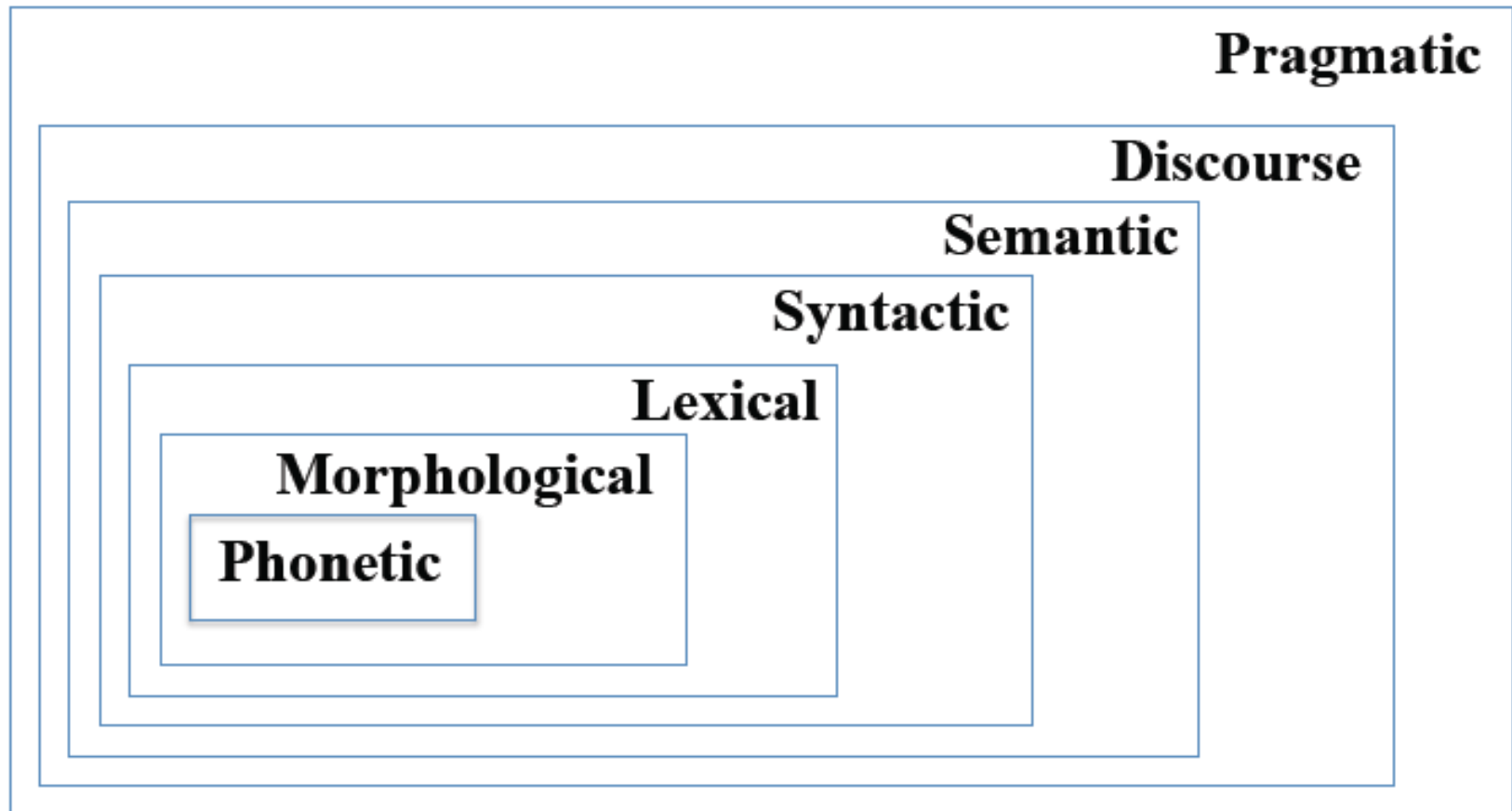
Discourse

- **Definition:** Analysis of language beyond a single sentence, focusing on how multiple sentences are connected to form a conversation.
- John bought a car. He loves it. (**coreference resolution.**)
- It was raining heavily. Therefore, the match was cancelled.

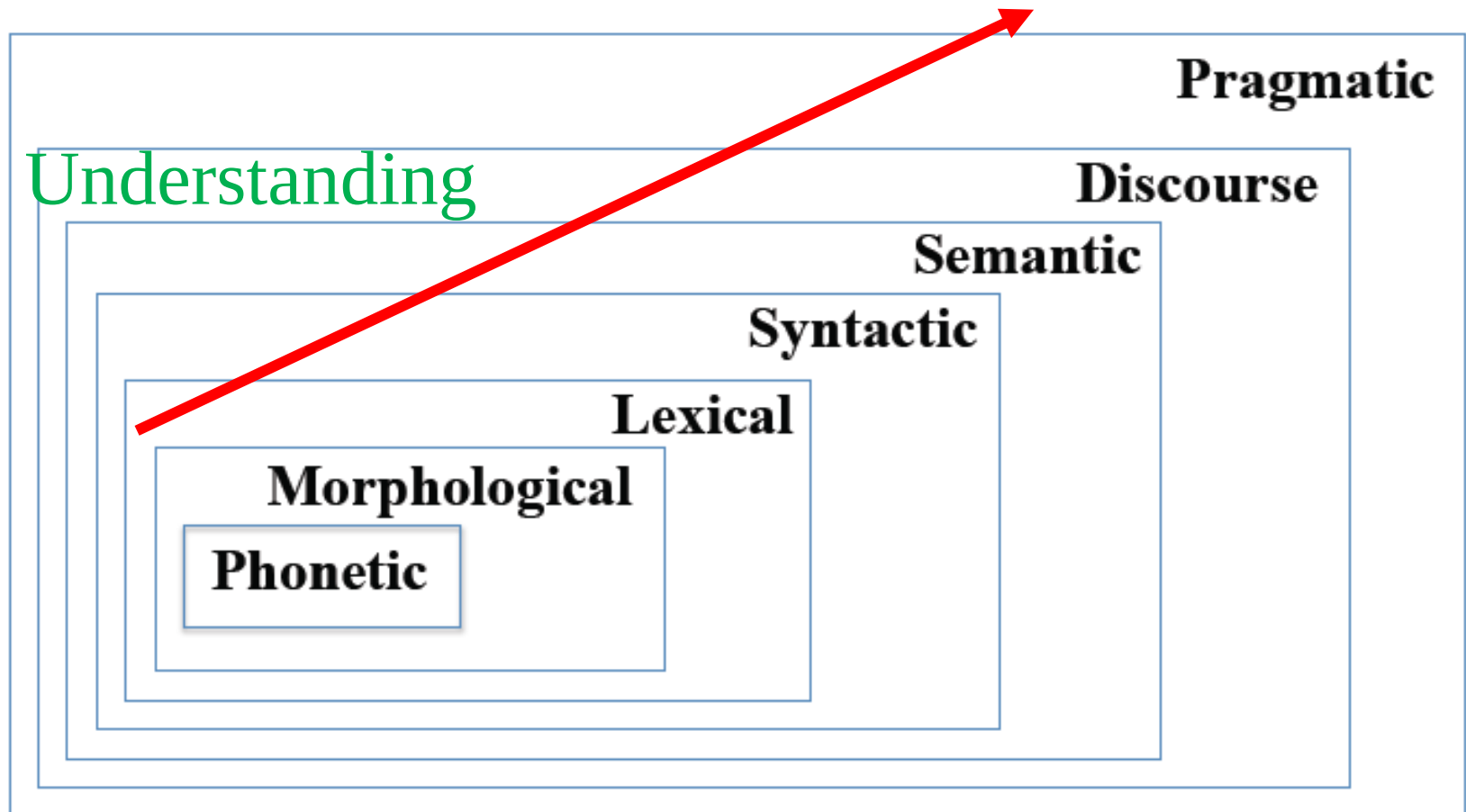
Pragmatics

- studies how context influences the meaning of language.
- Unlike lexical or compositional semantics, pragmatics considers:
 - speaker intention
 - situational context
 - shared knowledge
- Sentence
 - Can you open the window?
 - It's cold here.

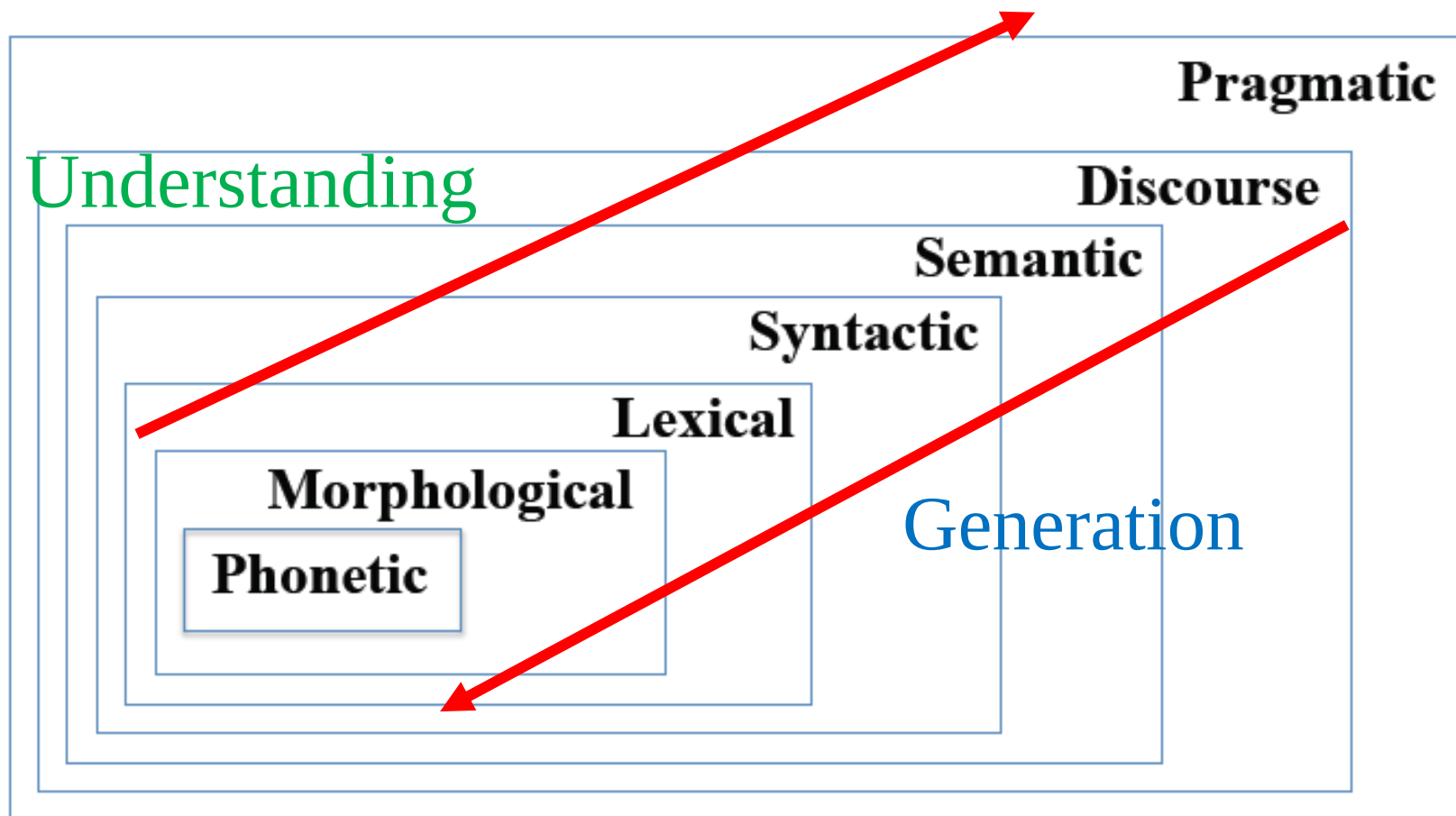
Synchronic Model of Language



Synchronic Model of Language



Synchronic Model of Language



Ultimate Goal

- ☐ Computational models of human language processing
 - ☐ Programs that operate internally the way humans do
- ☐ Computational models of human communication
 - ☐ Programs that interact like humans
- ☐ Computational systems that efficiently process text and speech

NLP Applications

- Machine Translation
- Information Retrieval
- Sentiment Analysis
- Information Extraction
- Question Answering
- Document Clustering
- Summarization
- Machine Aided Translation
- Text Segmentation
- Spelling and Grammar checking

Machine Translation

- Input:

Tout ce que vous produisez pour credit dans ce cours doit etre votre propre travail. Vous pouvez parler avec les autres etudiants (et les professeurs) de votre approche du probleme, mais ensuite vous devez rsoudre le probleme par vous-meme. Ce n'est pas seulement la facon la plus ethique d'apprendre le contenu de cette classe, mais aussi la plus efficace.

Machine Translation

- Output:

All that you produce for credit in this course must be your own work. You can speak with the other students (and the professors) about your approach about the problem, but then you must solve the same problem by you. It is not only the most ethical way to learn the contents from this class, but also most effective.

Machine Translation

- Original:

Everything you do for credit in this subject is supposed to be your own work. You can talk to other students (and instructors) about approaches to problems, but then you should sit down and do the problem yourself. This is not only the ethical way but also the only effective way of learning the material.

Information Extraction

10TH DEGREE is a full service advertising agency specializing in direct and interactive marketing. Located in Irvine CA, 10TH DEGREE is looking for an Assistant Account Manager to help manage and coordinate interactive marketing initiatives for a marquee automotive account. Experience in online marketing, automotive and/or the advertising field is a plus. Assistant Account Manager Responsibilities Ensures smooth implementation of programs and initiatives Helps manage the delivery of projects and key client deliverables ... Compensation: \$50,000-\$80,000 Hiring Organization: 10TH DEGREE



INDUSTRY	Advertising
POSITION	Assistant Account Manager
LOCATION	Irvine, CA
COMPANY	10TH DEGREE
SALARY	\$50,000-\$80,000

Information Extraction

- ▶ Goal: Map a document collection to structured database
- ▶ Motivation:
 - ▶ Complex searches (“Find me all the jobs in advertising paying at least \$50,000 in Boston”)
 - ▶ Statistical queries (“How has the number of jobs in accounting changed over the years?”)

Text Summarization

Agency Suspends Smallpox Vaccines for People With Heart Disease

Summary from the U.S.

A second health care worker has died of a heart attack (3) after receiving a smallpox vaccination (9) and officials are investigating whether vaccinations are to blame (3) for cardiac problems. (6) The vaccine never has been associated with heart trouble but as a precaution (3) the U.S. centers for Disease Control and Prevention (14) is advising people with a history of heart disease to be vaccinated (3) until further notice. (14) Strom suggested that the Bush administration reassess whether it necessary and safe to continue with its aggressive plan to inoculate millions of health care workers and emergency responders. (1)

Story keywords

vaccine, Heart, Smallpox, vaccinated, Disease

Source articles

1. [Vaccination program in peril after second death](#) (seattletimes.nwsource.com, 03/28/2003, 319 words)
2. [Wired News: Smallpox Shots: Proceed With Care](#) (Wired, 03/27/2003, 559 words)
3. [2nd worker dies after smallpox vaccination](#) (suntimes.com, 03/28/2003, 358 words)
4. [2nd worker dies after smallpox vaccine](#) (dallasnews.com, 03/28/2003, 499 words)
5. [Smallpox vaccine is reviewed after second fatal heart attack](#) (boston.com, 03/28/2003, 732 words)
6. [Second Smallpox Vaccine Death Evoked](#) (CBS News, 03/28/2003, 865 words)



ChatGPT

Who is the president of the United States?

The current President of the United States is Donald Trump. us

He became the **47th President of the United States** after winning the **2024 U.S. presidential election** and was inaugurated on **January 20, 2025**.

If you want, I can also explain:

- How the **U.S. presidential election system** works 🗳️
- The **roles and powers of the U.S. president**
- Or list **all recent U.S. presidents** for context.

ChatGPT

Who is the president of Russia?

The current President of Russia is Vladimir Putin. RU

He has served as President for multiple terms:

- 2000–2008 – President
- 2008–2012 – Prime Minister
- 2012–present – President again

In the 2024 Russian presidential election, Vladimir Putin was re-elected and began another term in May 2024.

If you want, I can also explain:

- The political system of Russia
- The difference between the President and Prime Minister in Russia.

ChatGPT

calculate $5 * 10 * 6 / 30$

The result of the expression $5 \times 10 \times 6 \div 30$ is 10.0. [↗]

ChatGPT

write the python code to add two numbers



Here is a simple Python code to add two numbers:

python

 Copy code

```
# Function to add two numbers
def add_numbers(num1, num2):
    return num1 + num2

# Example usage
number1 = 5
number2 = 10
result = add_numbers(number1, number2)
print(f"The sum of {number1} and {number2} is {result}")
```

ChatGPT

Yes. answer is correct . But the problem is he started the war before the other president. Then who is correct?

It seems you are comparing **two different times**:

1. Who is the president now
2. Who was the president when the war started

These can be different people.

1 When the war started

The large-scale war between Russia and Ukraine began with the Russian invasion of Ukraine (2022) on 24 February 2022.

At that time:

- President of Russia: Vladimir Putin
- President of the United States: Joe Biden

2 Who is the president now (2026)

Currently, the President of the United States is Donald Trump (after the 2024 election).

Dialogue Systems

User: I need a flight from Boston to Washington, arriving by 10 pm.

System: What day are you flying on?

User: Tomorrow

System: Returns a list of flights

Basic NLP Problems: Tagging

TAGGING: Strings to Tagged Sequences

a b e e a f h j $\Rightarrow$ a/C b/D e/C e/C a/D f/C h/D j/C

Example 1: Part-of-speech tagging

Profits/N soared/V at/P Boeing/N Co./N ,/,
easily/ADV topping/V forecasts/N on/P Wall/N
Street/N ./.

Example 2: Named Entity Recognition

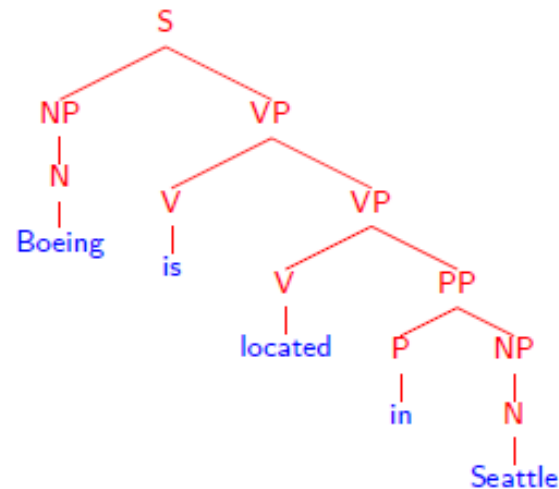
Profits/NA soared/NA at/NA Boeing/SC Co./CC
,/NA easily/NA topping/NA forecasts/NA on/NA
Wall/SL Street/CL ./.

Basic NLP Problems: Parsing

INPUT:

Boeing is located in Seattle.

OUTPUT:



Activity Time

- Tokenization:
 - Students can work individually or in small groups.
 - I can't believe U.S.A. won 3-0 today!

Activity Time

- Ambiguity (Write the meaning):
 - I saw the man with the telescope.
 - He went to the bank.
 - මම කොළ කඩ කඩ හිටියා

1. Lexical Ambiguity (Words with Multiple Meanings)

These sentences contain words that can be interpreted in different ways depending on the context.

- **Sentence:** කුම්බියා කුඹුරට ගියා.
 - **Translation:** "Koombiya (The ant / The person named Koombiya) went to the paddy field."
 - **Ambiguity:** "කුම්බියා" (Koombiya) can refer specifically to an ant, or it can be a colloquial nickname for a person.
- **Sentence:** ඇය බැංකුවට ගියා.
 - **Translation:** "She went to the bank."
 - **Ambiguity:** "බැංකුව" (Bankuwa) can mean a financial institution or a bench (loanword), though context usually clarifies this.
- **Sentence:** ඔහුගේ කතාව මට තේරුණේ නෑ.
 - **Translation:** "I didn't understand his story/speech."
 - **Ambiguity:** "කතාව" (Kathawa) can mean a story, a speech, or even a conversation. 

Why is NLP Hard?

[example from L.Lee]

“At last, a computer that understands you like your mother”

Ambiguity

“At last, a computer that understands you like your mother”

1. (*) It understands you as well as your mother understands you
2. It understands (that) you like your mother
3. It understands you as well as it understands your mother

1 and 3: Does this mean well, or poorly?

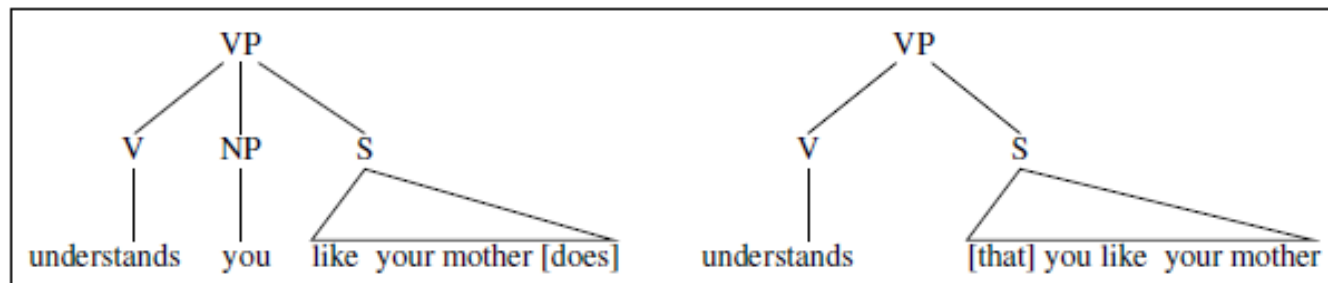
Ambiguity at Many Levels

At the **acoustic** level (speech recognition):

1. “ ... a computer that understands you **like your** mother”
2. “ ... a computer that understands you **lie cured** mother”

Ambiguity at Many Levels

At the **syntactic** level:



Different structures lead to different interpretations.

Ambiguity at Many Levels

At the **semantic** (meaning) level:

Two definitions of “mother”

- ▶ a woman who has given birth to a child
- ▶ a stringy slimy substance consisting of yeast cells and bacteria; is added to cider or wine to produce vinegar

This is an instance of **word sense ambiguity**

More Word Sense Ambiguity

At the **semantic** (meaning) level:

- ▶ They put money in the *bank*
= buried in mud?
- ▶ I saw her duck with a telescope

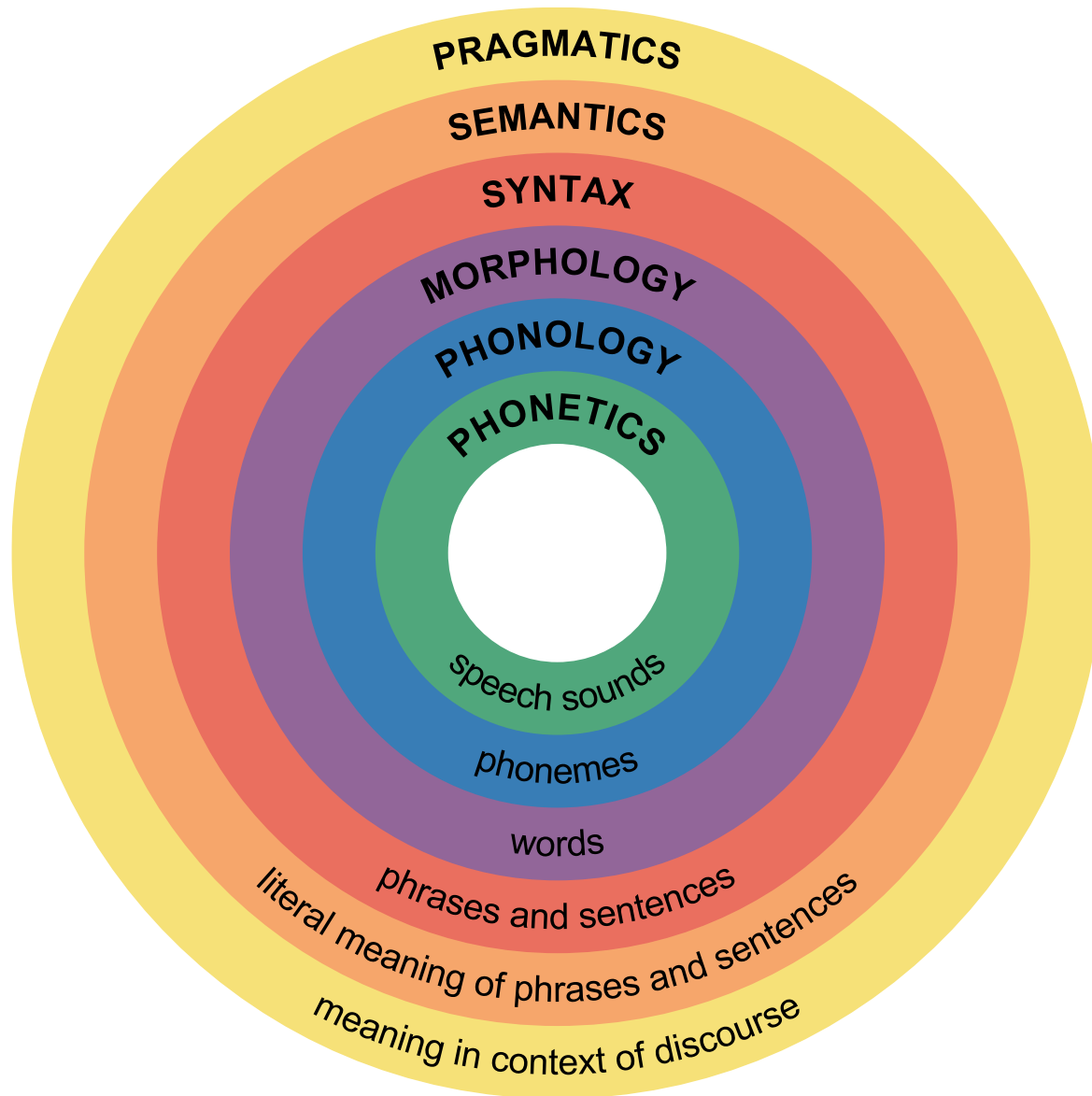
Ambiguity at Many Levels

At the **discourse** (multi-clause) level:

- ▶ Alice says they've built a computer that understands you like your mother
- ▶ But she ...
 - ... doesn't know any details
 - ... doesn't understand me at all

This is an instance of **anaphora**, where she co-refers to some other discourse entity

Major Levels of Linguistics



What is Unicode?

	0D8	0D9	0DA	0DB	0DC	0DD	0DE	0DF
0		ඌ	ඬ	ධ	ඳ	ඳ		
1		ඌ	ඌ	ඌ	ඌ	ඌ		
2	ඌ	ඌ	ඌ		ඌ	ඌ		ඌ
3	ඌ	ඌ	ඌ	ඌ	ඌ	ඌ		ඌ
4		ඌ	ඌ	ඌ	ඌ	ඌ		ඌ
5	ඌ	ඌ	ඌ	ඌ	ඌ			
6	ඌ	ඌ	ඌ	ඌ	ඌ	ඌ	ඌ	
7	ඌ		ඌ	ඌ			ඌ	

8	ඌ		ඌ	ඌ		ඌ	ඌ	
9	ඌ		ඌ	ඌ		ඌ	ඌ	
A	ඌ	ඌ	ඌ	ඌ	ඌ	ඌ	ඌ	
B	ඌ	ඌ	ඌ	ඌ		ඌ	ඌ	
C	ඌ	ඌ	ඌ			ඌ	ඌ	
D	ඌ	ඌ	ඌ	ඌ		ඌ	ඌ	
E	ඌ	ඌ	ඌ			ඌ	ඌ	
F	ඌ	ඌ	ඌ		ඌ	ඌ	ඌ	

